

Review.

Chapter 1.

§1.1. n -dimensional vector $V = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix}$ or $V = (V_1, V_2, \dots, V_n)$.

vector addition.

scalar multiplication.

Linear combination of vectors.

§1.2.

dot product or inner product.

$$V \cdot W = V_1 W_1 + V_2 W_2 + \dots + V_n W_n \rightarrow \text{scalar.}$$

Length $\|V\| = \sqrt{V \cdot V}$

Normalization $u = \frac{V}{\|V\|} \rightarrow \text{unit vector.}$

Angles between two vectors.

Cosine formula. $\cos \theta = \frac{V \cdot W}{\|V\| \|W\|} \Rightarrow V \cdot W = 0 \text{ if } V \perp W.$

Schwartz inequality $|V \cdot W| \leq \|V\| \|W\|.$

Chapter 2.

§2.1.

$$A = [a_1 \mid a_2 \mid \dots \mid a_n], \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\Rightarrow Ax = x_1 a_1 + x_2 a_2 + \dots + x_n a_n \rightarrow \text{multiplication by columns}$$

$$A = \begin{bmatrix} a^1 \\ a^2 \\ \vdots \\ a^n \end{bmatrix} \Rightarrow Ax = \begin{bmatrix} a^1 \cdot x \\ a^2 \cdot x \\ \vdots \\ a^n \cdot x \end{bmatrix} \rightarrow \text{multiplication by rows}$$

Identity matrix

$$I = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}_{n \times n}$$

General matrix. m rows, n columns

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \rightarrow A = (a_{ij})_{m \times n}.$$

Transformation of a system of linear equations
to $AX = b$ (matrix form).

§2.2.

Pivot = First nonzero in the row that does the elimination.

$$\underline{\text{Multiplier}} = \frac{(\text{entry to be eliminated})}{(\text{Pivot})}$$

Failure of Gaussian Elimination. $\rightarrow$ Three cases.

Gaussian elimination process.

§2.3.

Elimination matrix E_{ij} has an extra nonzero entry $-l_{ij}$ in the (i, j) -position of the Identity matrix.

$\Rightarrow E_{ij}A$ subtracts a multiple l_{ij} of row j from row i of A .

Associative Law $A(BC) = (AB)C$

Not commutative Law $AB \neq BA$ in general

Permutation matrix P_{ij} is obtained by exchanging row i and row j of the Identity matrix.

$\Rightarrow P_{ij} A$ exchanges row i and row j of A .

Gaussian elimination is equivalent to apply a sequence of Elimination matrices and permutation matrices to $[A \ b]$ (Augmented matrix) from the left of the linear system $AX=b$.

- Find corresponding E_{ij} 's, P_{ij} 's.

§2.4.

Matrix addition

Scalar multiplication of a matrix.

Matrix multiplication

$$A : m \times n \quad B : n \times p$$

$$\rightarrow C = AB : m \times p$$

Rule: The entry in row i and column j of AB is
(row i of A) $\cdot$ (row j of B)

$$\text{i.e., } C_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad 1 \leq i \leq m, \quad 1 \leq j \leq p.$$

Laws for matrix operations

$$A+B=B+A$$

$$\alpha(A+B) = \alpha A + \alpha B$$

$$A+(B+C) = (A+B)+C$$

$$C(AB) = CA+CB$$

$$(A+B)C = AC+BC$$

$$A(BC) = (AB)C$$

$AB \neq BA$ in general.

Power of matrix $A^P = A \cdot A \cdots A$ (P factors)

Block multiplication.

$$A = [a_1 \mid a_2 \mid \dots \mid a_n], \quad B = \begin{bmatrix} b^1 \\ b^2 \\ \vdots \\ b^n \end{bmatrix}$$

$$\Rightarrow AB = [a_1 b^1 + a_2 b^2 + \dots + a_n b^n]$$

§2.5.

A is invertible $\Leftrightarrow$ There exists a A^{-1} such that $AA^{-1} = I = A^{-1}A$.

Not all matrices have inverse.

$(A^{-1})^{-1} = A$ if A is invertible

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \text{if } ad-bc \neq 0$$

$$\begin{bmatrix} d_1 & & 0 \\ & d_2 & \\ 0 & & d_n \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{d_1} & & 0 \\ & \frac{1}{d_2} & \\ 0 & & \frac{1}{d_n} \end{bmatrix} \quad \text{if } d_i \neq 0 \text{ for } i=1, \dots, n$$

$(AB)^{-1} = B^{-1}A^{-1}$ if A and B are invertible.

$$(A_1 A_2 \dots A_m)^{-1} = A_m^{-1} \dots A_2^{-1} A_1^{-1}$$

Inverse of an Elimination matrix.

for example, $E_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ -e_{ij} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow E_{ij}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ e_{ij} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Gauss-Jordan Elimination for finding A^{-1}

$$[A \mid I] \longrightarrow [I \mid A^{-1}]$$

§2.6.

LU factorization $A = LU \rightarrow$ upper triangular.
 $\hookrightarrow$ lower triangular

3x3 matrix

$$L = \begin{bmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{bmatrix}$$

$U \rightarrow$ remaining upper triangular matrix after Gaussian elimination. $\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{bmatrix}$

LDU factorization

$$A = LDU \rightarrow \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{bmatrix} = \begin{bmatrix} d_1 & & \\ & d_2 & \\ & & d_3 \end{bmatrix} \begin{bmatrix} 1 & u_{12}/d_1 & u_{13}/d_1 \\ & 1 & u_{23}/d_2 \\ & & 1 \end{bmatrix}$$

$$\begin{aligned} Ax &= b \\ \Downarrow \\ (LU)x &= b \end{aligned} \Rightarrow \begin{cases} LY = b \rightarrow \text{using forward substitution to solve } Y \\ UX = Y \rightarrow \text{then using backward substitution to solve } X \end{cases}$$

§2.7.

$A^T \rightarrow$ Transpose of A

$$(A^T)_{ij} = A_{ji} \quad A: m \times n \rightarrow A^T: n \times m$$

$$\text{Rules: } (A+B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

$$(A^{-1})^T = (A^T)^{-1}$$

A is called a symmetric matrix iff $A^T = A$

The inverse of a symmetric matrix is also symmetric.

If $A = A^T$, $A = LDU$, then $U = L^T \Rightarrow A = LDL^T$.

Permutation matrix P has rows of I in any order.

P^T is also a permutation matrix $\left\{ \begin{array}{l} PA \rightarrow \text{row exchange of } A \\ AP \rightarrow \text{column exchange of } A \end{array} \right.$

$$P^{-1} = P^T$$

LU factorization with row exchange.